

Zeping Zhang

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RESEARCH INTERESTS

Artificial Intelligence; AI Security; Model Inversion Attacks; Privacy and Security of Multimodal Models; Multimodal Learning

ACADEMIC APPOINTMENTS

Jiangsu Second Normal University, School of Computer Engineering — Lecturer | Jan 2026–Present

Teach courses including Artificial Intelligence and Microcomputer Interface Technology; conduct research on artificial intelligence, AI security, and privacy issues in multimodal models. Recent research achievements includes a paper, a patent and a software copyright.

EDUCATION

Southeast University, School of Cyber Science and Engineering — Ph.D. in Cybersecurity | Sep 2020–Dec 2025

Direct-entry Ph.D. program; research focused on artificial intelligence, AI security, and privacy attacks & defenses of deep learning models.

Nanjing University of Posts and Telecommunications, Bell Honors School — B.Eng. in Communication Engineering | Sep 2016–Jun 2020

Bell Honors School is an honors school for cultivating top innovative undergraduate talent.

SELECTED PUBLICATIONS

- [1] **Z. Zhang**, X. Wang, J. Huang, and S. Zhang, "Analysis and Utilization of Hidden Information in Model Inversion Attacks," *IEEE Transactions on Information Forensics and Security*, 18, 4449–4462, 2023.
- [2] **Z. Zhang**, J. Huang, "Aligning the Domains in Cross-Domain Model Inversion Attack," *Neural Networks*, vol. 178, p. 106490, Oct. 2024.
- [3] **Z. Zhang**, J. Huang, "Exploiting the Connections Between Images and Deep Feature Vectors in Model Inversion Attacks," *Neurocomputing*, vol. 656, p. 131457, Dec. 2025.
- [4] **Z. Zhang**, J. Huang, and C. Ding, "Flexible Model Inversion Attack with soft biometric attribute reconstruction against face classifiers," *Journal of Systems Architecture*, vol. 175, p. 103739, Jun. 2026.
- [5] **Z. Zhang**, J. Zhang, H. Hu, and Y. Yin, "Metric Learning Based Open-Set Domain Adaptation Network," *Journal of Jiangsu Ocean University (Natural Science Edition)*, vol. 29, no. 04, pp. 39–46, 2020.
- [6] C. Liang, J. Huang, S. Zhang, and **Z. Zhang**, "MDV: Resolving the Auxiliary Data Dilemma in Model Extraction Defenses," *IEEE Transactions on Information Forensics and Security*, vol. 21, pp. 5360–5373, 2026.
- [7] Y. Zhang, J. Huang, **Z. Zhang**, S. Zhang, C. Ding, and X. Chen, "Consensus Learning with Multi-Party Perturbation Triggers for Secure Model Access," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 42, pp. 36084–36091, Mar. 2026.
- [8] Y. Zhang, J. Huang, P. Li, **Z. Zhang**, and C. Ding, "A Knowledge-Driven Dual Trigger Mechanism for Enhancing Model Security in Multi-Source Fusion Systems," *Knowledge-Based Systems*, vol. 335, p. 115183, Feb. 2026.
- [9] C. Ding, J. Huang, Q. Hao, **Z. Zhang**, and Y. Zhang, "Multimodal Privacy-Leaking Image Detection Method Based on Multi-Image Correlation," *Neurocomputing*, vol. 666, p. 132222, Feb. 2026.
- [10] Y. Zhang, J. Huang, **Z. Zhang**, P. Li, and C. Ding, "T-TNet: A Dual-Dependency Trigger Framework for Active Defense and Hierarchical Access Control via Multi-Domain Information Fusion," *Journal of Systems Architecture*, vol. 168, p. 103544, Nov. 2025.
- [11] C. Liang, J. Huang, **Z. Zhang**, and S. Zhang, "Defending Against Model Extraction Attacks with OOD Feature Learning and Decision Boundary Confusion," *Computers & Security*, vol. 136, p. 103563, Jan. 2024.
- [12] P. Li, J. Huang, H. Wu, **Z. Zhang**, and C. Qi, "SecureNet: Proactive Intellectual Property Protection and Model Security Defense for DNNs Based on Backdoor Learning," *Neural Networks*, vol. 174, p. 106199, Jun. 2024.
- [13] S. Zhang, J. Huang, **Z. Zhang**, P. Li, and C. Qi, "Compromise Privacy in Large-Batch Federated Learning via Model Poisoning," *Information Sciences*, vol. 647, p. 119421, Nov. 2023.
- [14] S. Zhang, J. Huang, **Z. Zhang**, and C. Qi, "Compromise Privacy in Large-Batch Federated Learning via Malicious Model Parameters," in *Algorithms and Architectures for Parallel Processing*, W. Meng, R. Lu, G. Min, and J. Vaidya, Eds., in Lecture Notes in Computer Science. Cham: Springer Nature Switzerland, 2023, pp. 63–80.

PATENTS

- [1] J. Huang, **Z. Zhang**, X. Wang, X. Zhang, and J. Luo, "Few-shot AOI Surface Defect Detection Method with Cross-Domain Transfer Capability," CN202210253629.1. Granted.
- [2] J. Huang and **Z. Zhang**, "Cross-Domain Black-Box Model Inversion Method," CN202211515085.8. Granted.
- [3] J. Huang and **Z. Zhang**, "Cross-Domain Model Inversion Method," CN202211515102.8.
- [4] J. Huang and **Z. Zhang**, "Few-shot Model Inversion Method Based on Deep Convolutional Feature Augmentation," CN202311791701.7.

RESEARCH PROJECTS

Student Innovation Funding Program, China Cyber Security Society | Oct 2022–Oct 2023 | Principal Investigator

Led a project on adversarial attacks and defenses in split learning. Focused on model inversion in split learning. The project was jointly funded by the Chinese Association of Cybersecurity and Ant Group; 240 students nationwide received funding in the program.

National Key R&D Program of China, "Key Technologies for Heterogeneous Object Resource Modeling and Exchange"

| Mar 2021–Mar 2023 | Research Participant

Participated in the project led by the Ph.D. advisor. Focused on the Middleware of heterogeneous IoT systems. Contributed to system design and research.

HONORS & AWARDS

- 2024 National Scholarship for Graduate Students, China
- 2024 Outstanding Graduate Student, Southeast University
- 2023 Advanced Individual for Academic Innovation, Southeast University
- 2024 Outstanding Academic Paper Award in Cyberspace Security, Jiangsu Province
- 2025 Third Prize, Outstanding Academic Paper Award in Cyberspace Security, Jiangsu Province
- 2021 National Second Prize, China Graduate Artificial Intelligence Innovation Competition
- 2021 National Third Prize, Huawei Cup China Graduate Mathematical Contest in Modeling

TEACHING & MENTORING

- Lecturer: Artificial Intelligence (Spring 2026); Microcomputer Interface Technology (Fall 2026), Jiangsu Second Normal University.
- Teaching Assistant: Pattern Recognition (Fall 2022), Southeast University.
- Undergraduate mentoring: supervised three undergraduate students on research of model inversion for their bachelor's thesis projects.
- Mentored undergraduate students in AI security project that received Third Prize at the Provincial Level in the China Collegiate Computer Design Competition.